

CSE331: Practice sheet-2 (CFG)

1. Give context-free grammars that generate the following languages

- a) $L = \{w \mid w \text{ contains at least three 1's}\}$.
- b) $L = \{w \mid w \text{ starts and ends with the same symbol}\}$.
- c) $L = \{w \mid \text{the length of } w \text{ is odd}\}$
- d) $L = \{w \mid \text{the length of } w \text{ is odd and its middle is 0}\}$
- e) $L = \{w \mid w = w^R, w \text{ is a palindrome}\}$
- f) $L = \{w \mid w = w^R \text{ AND } |w| \text{ is even, } w \text{ is a palindrome}\}$
- g) The empty set
- h) $L = \{w \mid w \text{ contains twice as many 1s as 0s}\}$
- i) $L = \{w \mid w \text{ contains more as than bs}\}$

2. Give a context-free grammar for each of the following languages.

- a) $L(G) = \{0^n 1^m 0^n \mid n, m \geq 0\}$ over the terminals $\{0, 1\}$
- b) $L(G) = \{a^n b^m c^k \mid n, m, k \geq 0 \text{ and } n = 2m + 3k\}$ over $\Sigma = \{a, b, c\}$
- c) $L(G) = \{0^n 1^m 0^m 1^n \mid n, m \geq 0\}$ over the terminals $\{0, 1\}$
- d) $L(G) = \{a^n b^m \mid 0 < n < m < 3n\}$. $\Sigma = \{a, b\}$
- e) $L(G) = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i=j \text{ or } i=k\}$. $\Sigma = \{a, b, c\}$
- f) $L(G) = \{a^i b^j c^k \mid i, j, k \geq 0 \text{ and } i=j \text{ or } j=k\}$. $\Sigma = \{a, b, c\}$
- g) $L(G) = \{a^n b^m c^m d^{2n} \mid n \geq 0, m > 0\}$.
- h) $L(G) = \text{set of all strings } w \text{ over } \{a, b\} \text{ such that } w \text{ is not a palindrome.}$

3. Consider the following context-free grammar $\Sigma = \{0, 1\}$.

$S \rightarrow A 1 B$
 $A \rightarrow 0A \mid \varepsilon$
 $B \rightarrow 0B \mid 1B \mid \varepsilon$

Give a) leftmost and b) rightmost derivations and c) parse tree for the following strings.

- a) 0010101
- b) 10100
- c) 00011

d) Explain/Prove why the grammar below is unambiguous.

4. Which language generates the grammar G given by the productions.

$S \rightarrow aSa \mid aBa$
 $B \rightarrow bB \mid b$

5. Explain/Prove why the grammar below is ambiguous.

$S \rightarrow 0A \mid 1B$
 $A \rightarrow 0AA \mid 1S \mid 1$
 $B \rightarrow 1BB \mid 0S \mid 0$

6. Given the following ambiguous context free grammar.

$$\begin{aligned} S &\rightarrow Ab \mid aaB \\ A &\rightarrow a \mid Aa \\ B &\rightarrow b \end{aligned}$$

- (a) Find leftmost and rightmost derivations for aaaaab, aabb, ab.
- (b) Show the parse trees for the above strings.
- (c) Is the above grammar ambiguous, give example?

7. Given the following ambiguous context free grammar.

$$\begin{aligned} S &\rightarrow Ab \mid AaB \\ A &\rightarrow a \mid Aa \\ B &\rightarrow b \end{aligned}$$

- (a) Find leftmost and rightmost derivations for aaaaab, aabb, ab.
- (b) Show the parse trees for the above strings.
- (c) Is the above grammar ambiguous, give example? Find an equivalent unambiguous context-free grammar.
- (d) Give the **unique** leftmost derivation and parse tree for the above strings generated from the unambiguous grammar above.

8. Show that the following grammar is ambiguous.

$$\begin{aligned} S &\rightarrow aEbS \\ S &\rightarrow aEbScS \mid \epsilon \\ E &\rightarrow d \end{aligned}$$

9. Consider the grammar with start symbol D, $\Sigma = \{c, a, b, ., 0, 1\}$

$$\begin{aligned} D &\rightarrow TL \\ T &\rightarrow c \mid Tc \\ L &\rightarrow L.V \mid V \\ V &\rightarrow a \mid b \mid 0 \mid 1 \mid Va \mid Vb \mid V0 \mid V1 \end{aligned}$$

- i) Derive ccab.01 using leftmost derivation
- ii) Derive cabb0011.ab1 using rightmost derivation

10. Show that the following grammar is ambiguous. [hint use string aab]

$$S \rightarrow aS \mid aSbS \mid \epsilon$$

- Give a)leftmost and b)rightmost derivations and c)parse tree for the following strings.
- d)Remove ambiguity of the above grammar.